



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : BIO F421
Course Title : ENZYMOLOGY
Instructor-in-Charge : P. R. DEEPA (deepa@pilani.bits-pilani.ac.in)
Instructor(s) : PANKAJ KUMAR SHARMA (pankajsharma@pilani.bits-pilani.ac.in)
Tutorial/Practical Instructors: -

1. Course Description:

Enzyme nomenclature and classification; isolation and purification; structures; kinetics; regulation of enzymatic reactions; evaluation of enzymes and other proteins.

2. Scope and Objective of the Course:

This course is designed to provide a broad introduction about enzymes, including their isolation, purification, structure and kinetics. Efforts will be made to understand the molecular basis of how enzymes catalyze the biological and non-natural reactions. Emphasis will be given to study the practical and industrial application of enzymes. This course integrates basics as well as current advances in enzymology and will create research interest among the students.

3. Text Books:

- 1). Trevor Palmer and Philip Bonner, *Enzymes: Biochemistry, Biotechnology, Clinical Chemistry.*, East-West Press Private Limited, New Delhi, 2nd Edition, 2008.

4. Reference Books:

- 1). Lehninger Principles of Biochemistry. David L. Nelson, Michael M. Cox. W.H. Freeman & Company, NY, USA, 7th Edition, 2017.
- 2). Methods in Enzymology - Guide to Protein Purification, Vol 463, Edited by BR Richard & Murray P. Deutscher. Academic Press, 2nd Edition, 2009.
- 3). Fermentation Microbiology and Biotechnology, Edited by El-Mansi EMT, Bryce CFA et al., CRC Press, Taylor & Francis Group, USA, 3rd Edition, 2012.
- 4). Peer-reviewed articles and relevant web-based resources will be discussed to enhance coverage of select topics.



5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
I Course & Class Introductions	L#1-2: Broad overview of the subject and field of Enzymology and Quick preview of the entire course contents.	Chap 1, T1	Getting introduced to the basic concept of enzymes, their significant properties, and varied applications in medicine, biotechnology and industry .
II Enzyme nomenclature and classification	L# 3-4: Brief history of Enzymology; EC system of classification; Main classes of enzymes, with examples.	Chap 1, T1	Will be familiarized with the scientific system of enzyme naming and the different types of enzymes.
III Enzyme structure, conformations, and specificity of enzyme actions	L# 5-7: Enzyme structure and conformations; Types of enzyme specificity; Specificity Hypotheses.	Chap 2, T1; Chap 4, T1	Will be able to appreciate the close association between enzyme protein structure and its biologic function.
IV Enzyme Kinetics & Mechanism of enzyme action and regulation	L# 8-11: Introduction to catalysis and kinetics; Kinetics of single-substrate enzyme-catalyzed reactions; Enzyme inhibition; Kinetics of multi-substrate enzyme-catalyzed reactions. Mechanisms of catalysis with examples; Enzymes and metabolic regulation; Investigation of active site structure.	Chap 6-9, T1; Chap 10, 11, 14, T1; R1	Will understand the different aspects of enzyme reaction kinetics, and mechanistic basis of bio-catalysis.
V Isolation and Purification	L# 12-19: Extraction of enzymes; Enzyme assay methods; Enzyme purification techniques.	Chap 15-16, T1; R2	Will be made familiar with analytical methods used in enzyme isolation and purification from cells.
VI Instrumental techniques in enzymatic analysis	L# 20-21: Various Instrumental techniques used in Enzymatic analysis	Chap 18, T1	Will learn about key techniques used in monitoring enzyme-catalyzed reactions.
VII Enzymes as Analytical Reagents	L# 22-23: Kinetic methods; Enzyme-immunoassays	Chap 17, T1	Will get to know of the applications of enzymes in analytical techniques.
VIII Immobilization of enzymes	L# 24-26: Methods of immobilization, Properties of immobilized enzymes and Applications	Chap 20, T1	Will learn of the need, methods and applications of enzyme immobilization.
IX Applications of enzymes	L# 27-35: Applications of enzymes and enzymatic analysis in clinical medicine, biotechnology, and pharmacology; Industrial Production of enzymes and applications (examples: in Food, Process industry).	Chap 19-21, T1	Will appreciate the diverse roles and applications of enzymes in clinical medicine, and several industries.



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X Recent advances and Innovations in Enzyme Technology	L# 36-38: Enzyme & Co-factor engineering – Current trends and Future prospects in Pharmaceutical & Fermentation Industries. Enzymes & Biosensors; Latest advances in enzymes - Industrial and Research-applications will be discussed.	R3; R4	Will be made aware of the unceasing technological potentials of enzymes in industry.
XI Course Summary	L# 39: Pointers for R&D and industrial applications of enzymes		Recapitulating the salient features of enzymes and their immense applications & potentials in medicine and industry.

Please note that a few expert guest lecture(s) would be arranged on specific themes.

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30	<TEST_1>	CB
Comprehensive Examination	3 h	40	<TEST_C>	CB / OB
Quiz(zes) / Assignment(s) / Mini-Project (including Presentation + Report)	Variable	30	Announced/surprise Details of Mini-Project will be discussed during class sessions	CB / OB

7. Chamber Consultation Hour: Will be announced in class. Prior Email communication from student to Instructor is preferred.

8. Notices: Announcements will be made during class sessions, and/ or shared by email.

9. Make-up Policy: Make-up may be granted only to genuine cases (hospitalization), subject to getting permission from IC. No make-up for Quiz/Assignments/Mini-projects under any circumstance.

10. Important Note:

- Regular attendance and active participation in class will be monitored and recorded. Absenting from classes without prior permission, lack of punctuality in reporting to class, non-submission of evaluative reports/assignments as per the scheduled time-line, can all impact Grades during evaluation.
- Mini-projects will be allotted only to the students present in the class during the project topic allotment and discussion session. In-absentia students will not be considered for mini-project evaluations.
- For open book evaluation components, only prescribed text book and self-prepared handwritten notes will be allowed.
- A student should obtain 30% of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

Instructor-in-charge
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